

Quiet Orbit QO-R1

Summary sheet. USB-C powered four-LED ambient lamp board, 57 × 57 mm.

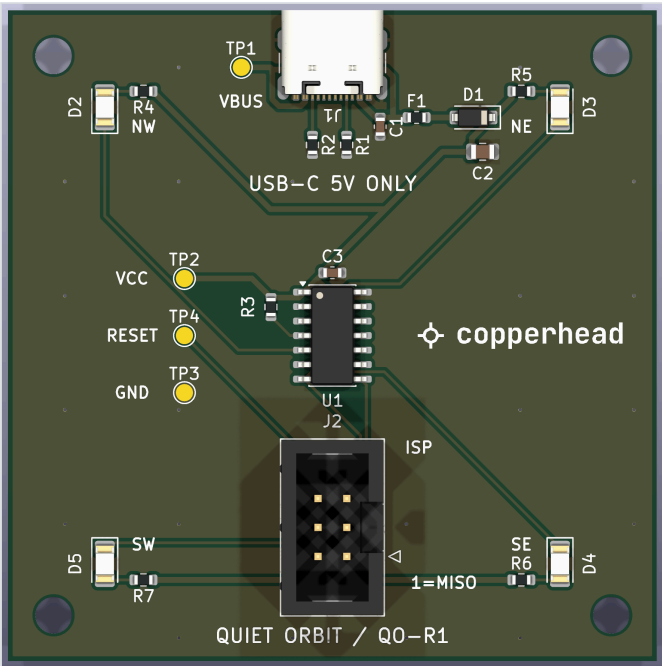
Design files dated 2026-09-21. For the full description see QO-R1-reference.pdf.

Status: not built, not tested.

- No board has been fabricated or powered.
- The firmware has not been compiled.
- Every part number in the BOM is unverified.
- D1 (1N4148W) is the wrong part. Fit a 1 A Schottky (B5819W or 1N5819HW class) in the same SOD-123 footprint.

Key figures

Board	57 × 57 mm, 2 layers, 1.6 mm, all SMT on top
Input	USB-C, 5 V sink only. 5.1 kΩ on CC1 and CC2. No data, no PD
Protection	100 mA PTC (F1), series diode (D1)
MCU	ATtiny84A, SOIC-14, internal 8 MHz RC
LEDs	4 × amber, 1206, 590 to 605 nm, active-low
LED current	2.26 to 4.38 mA each (calculated, 680 Ω)
PWM	4 hardware channels, 8-bit, about 488 Hz
Fade	Triangle, about 1.0 s period, channels 90° apart
Supply current	30.6 mA calculated maximum. Limit is 100 mA
LED positions	Corners of a 40 mm square, centred on the board
Mounting	4 × M3, 3.2 mm NPTH, 4 mm from each edge
Programming	2 × 3 AVR ISP header, 2.54 mm. Power from USB; ISP VCC pin is sense only
Enclosure	80 × 80 × 25 mm, four light windows



Top side. Render, not a photograph.

LED channels

Pos.	LED	R	U1 pin	PWM	Shared with
NW	D2	R4	5 (PB2)	OC0A	none
NE	D3	R5	6 (PA7)	OC0B	none
SE	D4	R6	7 (PA6)	OC1A	ISP MOSI
SW	D5	R7	8 (PA5)	OC1B	ISP MISO

SCK (pin 9) and RESET (pin 4) carry no LED. RESET has a 10 kΩ pull-up.

Board as committed

Footprints	27
Tracks	89 segments, 376.5 mm (320.8 top, 55.8 bottom)
Vias	24, 0.65 / 0.3 mm. 21 are GND stitching
Pours	GND, both sides
Rules	0.2 mm clearance. 0.25 mm signal, 0.5 mm power track. 0.4 mm copper to edge
ERC / DRC	0 / 0 violations, 0 unconnected, 0 parity issues (KiCad 10.0.4, all severities)

Bill of materials

Ref	Value	Package	MPN (unverified)	LCSC
J1	USB-C receptacle	GCT USB4105, 16 pin	USB4105-GF-A	C3020560
F1	PTC, 100 mA hold	0603	MF-FSMF010X-2	C210355
D1	1N4148W (replace with 1 A Schottky)	SOD-123	1N4148W-7-F	C81598
C1	1 μ F X7R	0603	CC0603KRX7R9BB105	none found
C2	4.7 μ F X7R	0805	GRM21BR71A475KA73L	none found
C3	100 nF X7R	0603	CC0603KRX7R9BB104	C14663
R1, R2	5.1 k Ω 1%	0603	RC0603FR-075K1L	C105580
R3	10 k Ω 1%	0603	RC0603FR-0710KL	C98220
R4 to R7	680 Ω 1%	0603	RC0603FR-07680RL	C112310
D2 to D5	Amber LED	1206	150120AS75000	C5478408
U1	ATTiny84A	SOIC-14	ATTINY84A-SSUR	C220597
J2	AVR ISP, 2 $\times$ 3, shrouded	2.54 mm, THT	61200621621	none found
TP1 to TP4	VBUS, VCC, GND, RESET	1.5 mm pad	bare pad	n/a

Check before ordering

- Replace D1 with a Schottky, as above. The board layout does not change.
- Verify every BOM line against a datasheet and a distributor. Three lines have no LCSC match.
- Open the gerbers in a viewer. Nobody has done this yet. No fab has run a DFM check.
- The J1 footprint puts GND pads 0.194 mm from its NPTH locating pegs. A fab with a 0.2 mm limit will query it.
- Placement rotations in the CPL file are KiCad's and have not been corrected for any assembler.
- Test points are marked DNP in docs/BOM.md but included in the BOM in the schematic. One of the two is wrong.
- The LED and M3 hole courtyards are 0.47 mm apart at each corner. An enclosure baffle has to fit around the screw head.